CLASS 10 · CBSE · NCERT MATHEMATICS · FORMULA BANK

Chapter 12: Surface Areas and Volumes

Every formula and result of the chapter on one sheet

1	Cuboid	TSA $= 2(lb + bh + hl)$; volume $= lbh$; diagonal $= \sqrt{l^2 + b^2 + h^2}$
2	Cube	TSA $= 6a^2$; volume $= a^3$; diagonal $= a\sqrt{3}$
3	Cylinder: curved surface area	$2\pi rh$
4	Cylinder: total surface area	$2\pi r(r + h)$
5	Cylinder: volume	$\pi r^2 h$
6	Cone: slant height	$l = \sqrt{r^2 + h^2}$
7	Cone: curved surface area	πrl
8	Cone: total surface area	$\pi r(l + r)$
9	Cone: volume	$\frac{1}{3}\pi r^2 h$
10	Sphere	Surface area $= 4\pi r^2$; volume $= \frac{4}{3}\pi r^3$
11	Hemisphere	CSA $= 2\pi r^2$; TSA $= 3\pi r^2$; volume $= \frac{2}{3}\pi r^3$
12	Cone on a hemisphere	Surface area $= \pi rl + 2\pi r^2$; volume $= \frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3$
13	Cylinder with two hemispherical ends	Surface area $= 2\pi rh + 4\pi r^2$; volume $= \pi r^2 h + \frac{4}{3}\pi r^3$
14	Hollow cylinder	Volume of material $= \pi(R^2 - r^2)h$
15	Units	1 litre $= 1000$ cm^3; 1 m^3 $= 1000$ litres